

Lakshya JEE 2.0 2025

Maths

DPP: 2

Method of Differentiation

Q1 If $y = \log(\sin(3x + 5))$, find $\frac{dy}{dx}$

- (A) $3 \cot(5x + 3)$
 (B) $3 \cot(3x + 5)$
 (C) $3 \tan(3x + 5)$
 (D) $3 \tan(5x + 3)$

Q2 If $y = \log(\sqrt{x-1} - \sqrt{x+1})$, find $\frac{dy}{dx}$

- (A) $\frac{-1}{2\sqrt{x-2}}$
 (B) $\frac{1}{2\sqrt{x^2-2}}$
 (C) $\frac{-1}{2\sqrt{x^2-1}}$
 (D) $\frac{1}{2\sqrt{x^2+2}}$

Q3 If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \infty}}}$,

then $\frac{dy}{dx}$ is:

- (A) $\frac{x}{2y-1}$
 (B) $\frac{x}{2y+1}$
 (C) $\frac{1}{x(2y-1)}$
 (D) $\frac{1}{x(1-2y)}$

Q4 If $f(x) = \frac{x-1}{x+1}$, find $\frac{d(f(f(f(x))))}{dx}$

- (A) $\frac{2}{(x-1)^2}$
 (B) $\frac{2}{(x-1)^3}$
 (C) $\frac{-2}{(x-1)^2}$
 (D) $\frac{-2}{(x-1)^3}$

Q5 If $f(x) = \log_x(\ln x)$, then $f'(x)$ at $x = e$ is

Q6 If $\log(x+y) = 2xy$, then $y'(0) =$

Q7 If $y\sqrt{1-x^2} + x\sqrt{1-y^2} = 1$, then $\frac{dy}{dx} =$

- (A) $-\sqrt{\frac{1-y^2}{1-x^2}}$
 (B) $\sqrt{\frac{1-y^2}{1-x^2}}$
 (C) $\sqrt{\frac{1-x^2}{1-y^2}}$
 (D) $-\sqrt{\frac{1-x^2}{1-y^2}}$

Q8 If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, then $(1-x^2) \frac{dy}{dx}$ is equal to:

- (A) $x + y$ (B) $1 + xy$
 (C) $1 - xy$ (D) $xy - 2$

Q9 If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$, then:

- (A) 1
 (B) -1
 (C) $\frac{1}{\sqrt{2}}$
 (D) None of these

Q10 $\frac{d}{dx} \cos^{-1} \sqrt{\cos x}$ is equal to:

- (A) $\frac{1}{2} \sqrt{1 + \sec x}$
 (B) $\sqrt{1 + \sec x}$
 (C) $-\frac{1}{2} \sqrt{1 + \sec x}$
 (D) $-\sqrt{1 + \sec x}$

Q11 If $y = \tan^{-1} \sqrt{\frac{1+\cos x}{1-\cos x}}$, where $0 < x < \pi$,

then $\frac{dy}{dx}$ is:

- (A) $\frac{-1}{2}$
 (B) 0
 (C) 1
 (D) -1

Q12



If

$$y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right], \left(0 < x < \frac{\pi}{2} \right)$$

then $\frac{dy}{dx} =$

- (A) $\frac{1}{2}$
 (B) $\frac{2}{3}x$
 (C) $3x$
 (D) 1

Q13 If $y = \cos^{-1} \left(\frac{5 \cos x - 12 \sin x}{13} \right)$, where

 $x \in \left(0, \frac{\pi}{2} \right)$, then $\frac{dy}{dx}$ is:

- (A) 1 (B) -1
 (C) 0 (D) none of these

Q14 Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2 \cos 2x \\ \cos 3x & \sin 3x & 2 \cos 3x \end{vmatrix}$. Then

find the value of $f' \left(\frac{\pi}{2} \right)$

- (A) 0 (B) 1
 (C) 2 (D) 3

Q15 If $y = (1+x)(1+x^2)(1+x^4)(1+x^8) \dots (1+x^{1006})$

then the value of $y'(0)$ is:

- (A) 503 (B) 0
 (C) 1 (D) 1006

Q16 If $x^y = e^{x-y}$, then $\frac{dy}{dx} =$

- (A) $\frac{(\log x - 1)}{(1 + \log x)^2}$
 (B) $\frac{\log x}{(1 + \log x)^2}$
 (C) $\frac{1}{(1 + \log x)^2}$
 (D) $-\frac{1}{(1 + \log x)^2}$

Q17 If $y = e^{x+e^{x+e^{x+\dots \text{to } \infty}}}$, then $\frac{dy}{dx} =$

- (A) $\frac{y}{1-y}$
 (B) $\frac{1-y}{y}$
 (C) $\frac{y}{1+y}$
 (D)

$$\frac{1+y}{y}$$

Q18 If $e^{x+y} - x = 0$, then $\frac{dy}{dx} =$

- (A) $\frac{x}{1-x}$
 (B) $\frac{x-1}{x}$
 (C) $\frac{1-x}{x}$
 (D) $\frac{x}{1+x}$

Q19 If

$$y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \text{to } \infty}}}$$

then $\frac{dy}{dx} =$

- (A) $\frac{\sin x}{2y-1}$
 (B) $\frac{\cos x}{2y-1}$
 (C) $\frac{2y-1}{\sin x}$
 (D) $-\frac{\sin x}{2y-1}$

Q20 Find $\frac{dy}{dx}$, when

$$x = a(t - \sin t), y = a(1 - \cos t)$$

- (A) $\sin \frac{t}{2}$
 (B) $\cos \frac{t}{2}$
 (C) $\cot \frac{t}{2}$
 (D) $\tan \frac{t}{2}$

Q21 If $x = a \left(t + \frac{1}{t} \right)$ and $y = a \left(t - \frac{1}{t} \right)$, find $\frac{dy}{dx}$

- (A) $\frac{y}{x}$
 (B) $\frac{x}{y}$
 (C) $-\frac{y}{x}$
 (D) $-\frac{x}{y}$

Q22 If $x = \sin^{-1} \left(\frac{3 \sin t + 4 \cos t}{5} \right)$ and

$$y = \sin^{-1} \left(\frac{6 \cos t + 8 \sin t}{10} \right)$$
, find $\frac{dy}{dx}$

- (A) 1
 (B) 0
 (C) -1
 (D) $\frac{1}{\sqrt{1-t^2}}$

Q23



If $x = \tan^{-1} \left(\frac{\sqrt{1+\sin t} + \sqrt{1-\sin t}}{\sqrt{1+\sin t} - \sqrt{1-\sin t}} \right)$ and
 $y = \tan^{-1} \left(\frac{\sqrt{1+t^2}-1}{t} \right)$, then $\frac{dy}{dx} =$

- (A) $\frac{1}{t^2-1}$
- (B) $t^2 + 1$
- (C) $\frac{1}{t^2+1}$
- (D) $t^2 - 1$

Q24 If

$$y = (1+x)(1+x^2)(1+x^4) \dots (1+x^{2n}),$$

then the value of $\left(\frac{dy}{dx} \right)_{x=0}$ is

- (A) 0
- (B) -1
- (C) 1
- (D) 2

Q25 If $f(x) = \log_x(\log_e x)$, then $f'(x)$ at $x = e$ is equal to

- (A) 1
- (B) 2
- (C) 0
- (D) $\frac{1}{e}$



Answer Key

Q1 (B)
Q2 (C)
Q3 (C)
Q4 (A)
Q5 $1/e$
Q6 1
Q7 (A)
Q8 (B)
Q9 (B)
Q10 (A)
Q11 (A)
Q12 (A)
Q13 (A)

Q14 (B)
Q15 (C)
Q16 (B)
Q17 (A)
Q18 (C)
Q19 (D)
Q20 (C)
Q21 (B)
Q22 (A)
Q23 (C)
Q24 (C)
Q25 (D)



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